

A conservative fix for constexpr uncaught_exceptions() and current_exception()

Abstract

The papers [P3818](#) and [P3820](#) explain the background of the problem we have making uncaught_exceptions() and current_exception(). In short, having those function constexpr is a breaking change in some cases.

Due to timing and lateness thereof, it's deemed prudent to just roll back adding constexpr to those functions, and figure out a solution in C++29 for allowing those functionalities in constant expression evaluation. That was the outcome of an LEWG discussion in a September 2025 teleconference.

Wording

In [exception.syn], remove the constexpr decl-specifiers from these functions:

```
constexpr int uncaught_exceptions() noexcept;  
using exception_ptr = unspecified ;  
constexpr exception_ptr current_exception() noexcept;
```

Before [uncaught.exceptions]/1, remove the constexpr decl-specifier:

```
constexpr int uncaught_exceptions() noexcept;
```

Before [propagation]/9, remove the constexpr decl-specifier:

```
constexpr exception_ptr current_exception() noexcept;
```

In [propagation]/12, edit as follows:

```
Effects: Creates an exception_ptr object that refers to a copy of e, as if:  
—try {  
—    throw e;  
—} catch(...) {  
—    return current_exception();  
—}  
Returns: An exception_ptr object that refers to a copy of e.
```

Strike the note In [propagation]/13:

```
[Note 5: This function is provided for convenience and efficiency reasons. —end note]
```